A Neural Dirichlet-Neumann Approach for Learning Euler Equation Flows

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Abstract

Abstract to be completed.

Keywords: Water waves; Dirichlet–Neumann operator; neural operators

1 Introduction

More needs to be added to introduction.

Governing Equations & Hamiltonian Formulation Obviously we should refine this portion later. We consider the evolution of a free surface on top of a two-dimensional fluid under the influence of gravity. We let x denote our horizontal direction, and y denote our vertical direction, and t denote time. We assume periodic boundary conditions on the horizontal, with domain length $L > 0$. Let $y = \eta(x, t)$ represent our free surface. We fix an impermeable bottom at a constant depth $y = -h$. We assume the fluid is incompressible, inviscid, and that its flow is irrotational, so that the fluid velocity is given by $u = \nabla\varphi$, and its velocity potential φ satisfies Laplace's equation:

$$\Delta\varphi = 0, \tag{1}$$

in the fluid region bounded below by the bottom at $y = -h$, and above by the free surface defined by $\eta(x, t)$. We assume the boundary conditions

$$\begin{aligned} \varphi_y &= 0, \quad \text{on } y = -h, \\ \eta_t + \nabla_x\eta \cdot \nabla_x\varphi - \varphi_y &= 0, \quad \text{on } y = \eta, \\ \varphi_t + \frac{1}{2}|\nabla\varphi|^2 + g\eta &= 0, \quad \text{on } y = \eta, \end{aligned} \tag{2}$$

where g denotes the acceleration due to gravity. We reduce the dimensionality of the classical potential flow formulation of the Euler equation problem by introducing the Dirichlet–Neumann operator (DNO), which takes Dirichlet data $\xi(x, t) = \varphi(\eta(x, t), t)$ and gives the corresponding Neumann data, i.e., the normal velocity at the free surface. The DNO G satisfies

$$G(\eta)\xi = (-\nabla_x\eta, 1)^\top \cdot \nabla\varphi|_{y=\eta}.$$

The system is Hamiltonian in η, ξ, with

$$H(\eta, \xi) = \frac{1}{2} \int_{\Omega} [\xi G(\eta)\xi + g\eta^2], dx. \tag{3}$$

Using the fact that

$$\eta_t = \frac{\delta H}{\delta \xi}, \qquad \xi_t = -\frac{\delta H}{\delta \eta}, \tag{4}$$

we can write the system (2) as

$$\begin{aligned} \eta_t &= G(\eta)\xi, \\ \xi_t &= -g\eta - \frac{1}{2}\xi_x^2 + \frac{(G(\eta)\xi + \eta_x \xi_x)^2}{2(1+\eta_x^2)}. \end{aligned} \tag{5}$$

Throughout this manuscript, we note that the operator $G(\eta)$ implicitly depends on h. When it is noteworthy, we note this dependence by writing the DNO as $G(\eta; h)$. For each admissible surface η, the operator $G(\eta)$ is linear in its Dirichlet argument ξ, self-adjoint, and positive semidefinite; it is positive definite after restriction to an appropriate mean-zero space. The Hamiltonian H of our system is invariant to time. In addition to being vital as a stability check for our classical algorithms, we will make use of energy conservation to increase the stability of our neural network models.

1.1 Related Work

1.2 Contribution

The primary objective of this work is to construct a neural approximation of the water-wave DNO whose evaluation is independent of the Craig–Sulem truncation order M, while retaining properties needed for reliable time integration. In particular, the proposed method is designed to achieve accurate DNO evaluation across several challenging solution classes, approximate Hamiltonian conservation, structural consistency, and stable long-time trajectories. The intended principal comparison is with a vanilla FNO trained only with the data-inconsistency loss, for which small one-step prediction errors need not imply structure-preserving or stable dynamics. The comparison with a vanilla FNO remains to be completed. Computational cost is examined as a practical consequence of the M-independent representation, rather than as the principal contribution.

2 Problem Setup

We now describe methods for solving the water wave equations in terms of solving for the DNO.

2.1 Forward Problem Solver

Craig–Sulem Series Add citations later. It is known that if η is Lipschitz, then G is analytic in η, hence G has a convergent Taylor series. We write

$$G(\eta) = \sum_{i \geq 0} G_i(\eta), \tag{6}$$

where G_i is the ith homogeneous term in the Taylor expansion of G. It can be shown that each term G_i can be calculated using an explicit recursion formula with base case $G_0 = |D_x| \tanh(h|D_x|)$, where $D_x = -i\nabla_x$, and i is the imaginary unit. We note that G_0 is thus independent of the surface shape η. By use of dynamic programming and the fast Fourier transform, it can be shown that the time complexity of calculating the Mth polynomial G_M is $\Theta(M^2 N \log N)$, where N is the spatial discretization dimension. Xu and Guyenne make use of this recursive method of calculating G up to the Mth Taylor term in [cite] to solve the water wave problem. The details of this algorithm are in the appendix.

Numerical Methods for CS Series Add typical method for calculating the DNO. Add challenges of calculating DNO.

Time Integrator Following Xu and Guyenne, we employ an implicit symplectic scheme to solve our time stepping PDE problem. Details in appendix. Sam

2.2 Learning Problem

Describe learning the DNO operator using a parametrized model, and a dataset $\mathcal{D}$ and loss $\mathcal{L}$. Need to emphasize that the target operator is depth dependent and the PDE is time dependent. Want to learn many classes of initial conditions and maintain long-time stability.

Deep Learning / Scientific Machine Learning Deep learning methods have recently seen success in being incorporated into solutions for scientific problems, especially those governed by partial differential equations. This has given rise to the field of Scientific Machine Learning. Practitioners leverage the well-known universal approximation results of deep neural networks to model unknown target functions using data. When extended to the appropriate domains and function spaces of partial differential equation solutions, this is often referred to as operator learning. Given the goal of replacing the costly computation of the DNO with a cheap-to-evaluate and accurate neural surrogate, we require describing a parametric neural network model G_θ, a dataset $\mathcal{D}$ to learn from, and a loss function $\mathcal{L}$ to enforce consistency of our parametric model with the target operator G. We describe our parametric family of models in Section 3.5, the construction of $\mathcal{D}$ in Section 4.1, and our choice of $\mathcal{L}$ in Section 3.1.

Operator Learning Operator learning is broadly characterized by methods for approximating mappings between Hilbert or Banach spaces. In practice, this is achieved by designing methods of approximation which are invariant to choice of discretization of the domain Ω. We achieve this by using a Fourier Neural Operator (FNO) backbone.

3 Proposed Method

We propose a method for solving the water wave equations by approximating the DNO. Additionally, we demonstrate that simply approximating the DNO pointwise is insufficient for maintaining stability in the downstream numerical PDE task. Thus we provide several physics-informed methods for ensuring that the learned DNO provides the necessary stability for PDE simulation.

3.1 Designing a Loss Function

We teach our model the law of the DNO operator by penalizing the following loss:

$$\mathcal{L}_{\text{DNO}} = \alpha_1 \mathcal{L}_{L^2} + \alpha_2 \mathcal{L}_{\text{MB}} + \alpha_3 \mathcal{L}_{\text{TG}} + \alpha_4 \mathcal{L}_{\text{HD}}, \tag{7}$$

where the coefficients α_i are a convex combination determined through hyperparameter search. The loss $\mathcal{L}_{L^2}$ is the ordinary L^2 loss, calculated spectrally. KC: New notations should be defined. The last three terms of our aggregate loss are each derived from the Hamiltonian of our system, helping ensure the learned dynamics of our network are consistent with physics. Each loss is carefully ablated.

3.2 Hamiltonian Mode-Balancing Penalty

Need to numerically show that during water wave simulation it is possible that certain modes tend to zero or certain modes completely dominate the output. As we are learning our operator chiefly in Fourier space, and using a spectral solver in the downstream PDE simulation, we aim to ensure that our learned operator respects the evolution of each active Fourier mode in a given instantiation of our time-stepping PDE. A key challenge in ensuring this is disallowing certain modes to dominate the evolution. Add supporting references. We show that a single-step loss which penalizes long-term mismatches in Fourier mode activity can be derived from the Hamiltonian of our problem. We begin by deriving a quadratic approximation to our Hamiltonian (3). Define $F(\varepsilon) := G(\varepsilon\eta)\xi$, for fixed η, ξ, and $\varepsilon > 0$. Applying a Taylor expansion on F about zero gives

$$F(\varepsilon) = F(0) + \varepsilon F'(0) + \mathcal{O}(\varepsilon^2). \tag{8}$$

Perturbing (3) by ε,

$$H(\varepsilon\eta, \varepsilon\xi) = \frac{1}{2}\int_\Omega \varepsilon\xi \cdot G(\varepsilon\eta)\varepsilon\xi \, dx + \frac{g}{2}\int_\Omega \varepsilon^2\eta^2 \, dx \tag{9}$$

$$= \frac{1}{2}\int_\Omega \varepsilon^2\xi[F(0) + \varepsilon F'(0) + \mathcal{O}(\varepsilon^2)]\xi \, dx + \frac{g}{2}\int_\Omega \varepsilon^2\eta^2 \, dx \tag{10}$$

$$= \varepsilon^2\left(\frac{1}{2}\int_\Omega \xi F(0) + \frac{g}{2}\int_\Omega \eta^2 \, dx\right) + \varepsilon^3\left(\frac{1}{2}\int_\Omega F'(0)\right) + \mathcal{O}(\varepsilon^4). \tag{11}$$

Therefore the quadratic Hamiltonian is

$$H_2(\eta, \xi) = \frac{1}{2}\int_\Omega [\xi G(0)\xi + g\eta^2], dx. \tag{12}$$

The Fourier symbol of the flat-surface DNO, $G(0)$, has known form:

$$\widehat{G(0)\xi}_k = |k| \tanh(h|k|)\widehat{\xi}_k. \tag{13}$$

Thus by using Parseval's theorem, one may calculate that for any η, ξ, $H_2(\eta, \xi) = \sum_k H_2(\eta, \xi)_k$, where

$$H_2(\eta, \xi)_k = \frac{1}{2}\left(|k| \tanh(h|k|)|\widehat{\xi}_k|^2 + g|\widehat{\eta}_k|^2\right). \tag{14}$$

Plugging (12) into (4) yields the linearized system

$$\eta_t^{\mathrm{L}} = G(0)\xi, \qquad \xi_t^{\mathrm{L}} = -g\eta. \tag{15}$$

Taking the Fourier transform yields, for nonzero modes k,

$$\partial_t\widehat{\eta_k^{\mathrm{L}}} = \widehat{G(0)\xi}_k, \qquad \partial_t\widehat{\xi_k^{\mathrm{L}}} = -g\widehat{\eta}_k. \tag{16}$$

Differentiating the first equation once more in time,

$$\partial_{tt}\widehat{\eta_k^{\mathrm{L}}} = \widehat{G(0)}_k\partial_t\widehat{\xi}_k \tag{17}$$

$$= -g\widehat{G(0)}_k\widehat{\eta_k^{\mathrm{L}}}. \tag{18}$$

Define $\omega_k^2 := g\widehat{G(0)}_k$, then note that

$$\partial_{tt}\widehat{\eta_k^{\mathrm{L}}} + \omega_k^2\widehat{\eta_k^{\mathrm{L}}} = 0 \tag{19}$$

is a harmonic oscillator with frequency ω_k, whose amplitude is

$$A_k^2 = |\widehat{\eta_k^{\mathrm{L}}}|^2 + \frac{|\partial_t\widehat{\eta_k^{\mathrm{L}}}|^2}{\omega_k^2}. \tag{20}$$

Furthermore, this quantity is invariant to time:

$$\begin{aligned}
\frac{d}{dt}A_k^2 &= \partial_t\widehat{\eta}_k^{\mathrm{L}}\,\overline{\widehat{\eta}_k^{\mathrm{L}}} + \widehat{\eta}_k^{\mathrm{L}}\,\partial_t\overline{\widehat{\eta}_k^{\mathrm{L}}} \\
&\quad + \frac{1}{\omega_k^2}\left(\partial_{tt}\widehat{\eta}_k^{\mathrm{L}}\,\partial_t\overline{\widehat{\eta}_k^{\mathrm{L}}} + \partial_t\widehat{\eta}_k^{\mathrm{L}}\,\partial_{tt}\overline{\widehat{\eta}_k^{\mathrm{L}}}\right) \\
&= \partial_t\widehat{\eta}_k^{\mathrm{L}}\,\overline{\widehat{\eta}_k^{\mathrm{L}}} + \widehat{\eta}_k^{\mathrm{L}}\,\partial_t\overline{\widehat{\eta}_k^{\mathrm{L}}} \\
&\quad + \frac{1}{\omega_k^2}\left(\omega_k^2\widehat{\eta}_k^{\mathrm{L}}\,\partial_t\overline{\widehat{\eta}_k^{\mathrm{L}}} + \omega_k^2\partial_t\widehat{\eta}_k^{\mathrm{L}}\,\overline{\widehat{\eta}_k^{\mathrm{L}}}\right) \\
&= 0.
\end{aligned}$$

We use this quantity to quantify the scale of the kth Fourier mode of the target $G(\eta)\xi$, for fixed η, ξ.

3.3 Tangent-Gain Penalty

For fixed η, ξ, write the error of our model G_θ against G by $\mathcal{E} = G_\theta(\eta)\xi - G(\eta)\xi$. If we assume that for some $c > 0$ our wave has a transporting profile, $G(\eta)\xi := -c\eta_x$, then, assuming $G_\theta(\eta)\xi \approx (1+\varepsilon)G(\eta)\xi$, we find $\mathcal{E} \approx -\varepsilon c\eta_x$, which is an accelerative error in the translating direction. We empirically find that when simulating the water wave equations using a naïve neural surrogate with certain initial wave profiles, translating error dominates the error accumulated. Add a small numerical experiment to support this. We thus provide a simple loss which targets general potential translation error accumulated during the simulation of the water wave PDE. For fixed x, η, and $a > 0$, the velocity of the translating operator $T_a(\eta) := \eta(x-a)$ is $-\eta_x$. Let P_{η_x} be the L^2 projection operator onto $\mathrm{span}\{\eta_x\}$. As before, let $\mathcal{E} = G_\theta(\eta)\xi - G(\eta)\xi$. We would like to penalize

$$\frac{\|P_{\eta_x}\mathcal{E}\|^2}{\|\eta\|_2^2} = \frac{|\langle \mathcal{E}, \eta_x\rangle|^2}{\|\eta\|_2^2\|\eta_x\|_2^2}, \tag{21}$$

where we divide by the square norm of η as is practice in deep learning to maintain scale invariance of our loss function. In practice, a target wave profile may have more than one translating component, with varying speeds, which motivates deriving a localized version of (21). We first define the periodic Gaussian kernel

$$K_\sigma(x-y) := \frac{1}{\sqrt{2\pi\sigma^2}} \sum_{m\in\mathbb{Z}} \exp\left(-\frac{(x-y+mL)^2}{2\sigma^2}\right),$$

for $x, y \in \mathbb{T}_L$ and $\sigma^2 > 0$ the variance/localization parameter of the kernel. We then define the following averaging operator:

$$(B_\sigma f)(x) := \int_0^L K_\sigma(x-y) f(y)\, dy.$$

For $k_n = 2\pi n/L$, the nth Fourier series coefficient of our kernel is $\widehat{K}_{\sigma,n} = L^{-1}\exp(-\sigma^2 k_n^2/2)$. For a periodic function $f(x) = \sum_{n\in\mathbb{Z}} \widehat{f}_n \exp(ik_n x)$, by the convolution theorem,

$$\widehat{(B_\sigma f)}_n = \exp(-\sigma^2 k_n^2/2)\widehat{f}_n.$$

The above Fourier multiplier form of B_σ is useful for us as numerically we calculate this convolution spectrally rather than physically. Upon defining our averaging operator, we replace our L^2 inner product with our localized inner product $\langle f, g\rangle_{\sigma,x} = \int_{\mathbb{T}_L} K_\sigma(x,y) f(y) g(y)\, dy$. We repeat the above step of projecting the error $\mathcal{E}$ onto the span of η_x under our new weighted inner product using the projection operator $P_{\eta_x}^{\sigma,x_0}$, where we emphasize the dependence on the fixed reference point x_0:

$$\frac{\|P_{\eta_x}^{\sigma,x_0}\mathcal{E}\|_{\sigma,x_0}^2}{\|\eta\|_2^2} = \frac{|\langle \mathcal{E}, \eta_x\rangle_{\sigma,x_0}|^2}{\|\eta\|_2^2\|\eta_x\|_{\sigma,x_0}^2}.$$

Lastly, we integrate over all reference points x_0 to define our loss $\mathcal{L}_{\mathrm{TG}}$:

$$\mathcal{L}_{\mathrm{TG}} = \frac{1}{\|\eta\|_2^2}\int_{\mathbb{T}_L} \|P_{\eta_x}^{\sigma,x_0}\mathcal{E}\|_{\sigma,x_0}^2\, dx_0. \tag{22}$$

3.4 Hadamard Derivative

Jonathan Intuitively learning the DNO pointwise from data is insufficient to guarantee stability during simulation of the time-stepping PDE. Notably, as our network only claims to be an approximation to the true DNO, even small numerical error in its approximation will force the network to make predictions on future states which may only necessarily be near the true PDE trajectory. We cite the following theorem of Lannes which can help encourage our network to be locally accurate in an appropriate manner.

Proposition 1. *Let η, ξ, and ζ be sufficiently smooth, and assume that the fluid depth remains strictly positive. Define*

$$q = G(\eta)\xi, \qquad B = \frac{q + \eta_x\xi_x}{1+\eta_x^2}, \qquad V = \xi_x - \eta_x B.$$

Then the map $\eta \mapsto G(\eta)\xi$ is Fréchet differentiable and

$$D_\eta G(\eta)[\zeta]\xi = -G(\eta)(\zeta B) - \partial_x(\zeta V).$$

A proof is provided in the appendix, as well as citations. To enforce local correctness of our learned model G_θ, we define the residual of G_θ with Proposition 1:

$$\mathcal{H}_\theta(\eta, \xi; \zeta) = D_\eta G_\theta(\eta)[\zeta]\xi + G_\theta(\eta)(\zeta B_\theta) + \partial_x(\zeta V_\theta),$$

where

$$B_\theta = \frac{G_\theta(\eta)\xi + \eta_x \xi_x}{1 + \eta_x^2}, \qquad V_\theta = \xi_x - \eta_x B_\theta.$$

We approximate this Fréchet derivative by sampling a random direction ζ on the unit sphere S^{d-1}, where d is our discretization dimension, and sampling $\varepsilon \sim \mathcal{U}(10^{-3}, 3 \times 10^{-3})$, then we calculate

$$D_\eta G_\theta(\eta)[\zeta]\xi \approx \frac{G_\theta(\eta + \varepsilon\zeta)\xi - G_\theta(\eta)\xi}{\varepsilon}.$$

The terms $\partial_x(\zeta V_\theta)$, η_x, and ξ_x are calculated by spectral differentiation. We then define the loss $\mathcal{L}_{\text{HD}}$ as

$$\mathcal{L}_{\text{HD}} = \mathbb{E}_{\substack{\zeta \sim \mathcal{U}(S^{d-1}) \\ \varepsilon \sim \mathcal{U}(10^{-3}, 3\times 10^{-3})}} [\mathcal{H}_\theta(\eta, \xi; \zeta)]. \tag{23}$$

3.5 Architecture

We now describe the architectural decisions we made in defining our family of parametric models for learning the DNO. Our network is built chiefly in Fourier space, as a composition of learned Fourier multipliers. We show that this allows our network to satisfy several properties of the DNO by construction, as well as enabling the discretization invariance trait which is desirable in operator learning.

CS Truncation The first term of (6) has a known Fourier symbol:

$$\widehat{G_0\xi}_k = |k| \tanh(h|k|)\widehat{\xi}_k. \tag{24}$$

The second term of (6), through the known Craig–Sulem recursion, is written

$$G_1(\eta)\xi = D \cdot (\eta D\xi) - G_0(\eta G_0 \xi). \tag{25}$$

We thus propose to learn the following approximation:

$$G(\eta)\xi \approx G_0\xi + G_1(\eta)\xi + G_\theta(\eta)\xi, \tag{26}$$

where G_θ is our learned surrogate. We should quantify the actual error between G and $G_0 + G_1$ to give some motivation. It is of course $\mathcal{O}(\eta^2)$.

Low Rankedness of Residual Jonathan

Depth and Wavenumber Conditioning As mentioned in Section 1, the DNO depends on the geometry of the prescribed water wave problem. In our two-dimensional fluid setting, this geometry dependence is explicit in the depth of our impermeable bottom at depth $-h$. In addition, due to the depth dependence in the base case of our CS recursion (24), the depth parameter h acts mode-by-mode on the DNO. Jonathan

Positive Definiteness and Linearity As mentioned in Section 1, for each admissible η the target operator $G(\eta)$ is linear in ξ and positive semidefinite, which motivates the following architectural decisions for the neural surrogate G_θ.

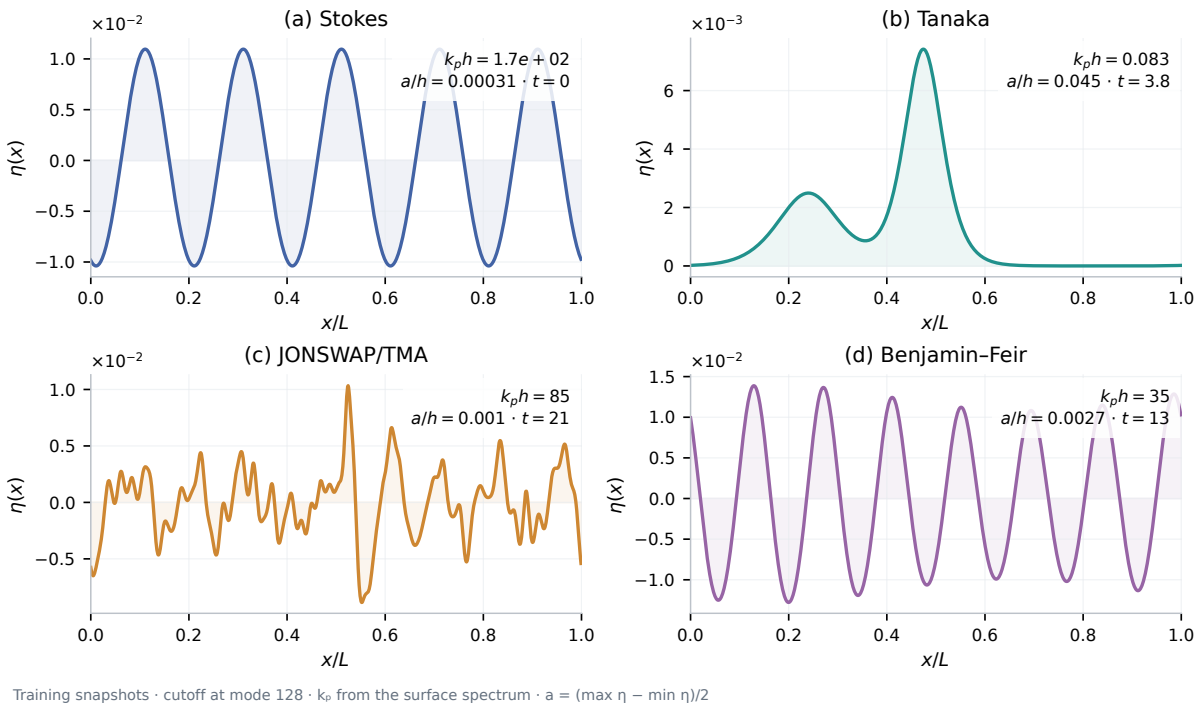


Figure 1: Training examples from the four wave families, selected near each family's median sampled DNO error. Here k_p is the dominant nonzero surface wavenumber and $a = (\max \eta - \min \eta)/2$.

3.6 Training the Neural Operator

Jonathan

4 Numerical Experiments

We compare the learned DNO with the Craig–Sulem (CS) reference solver. Here $G_\theta(\eta; h)\xi$ includes both the analytic terms and the learned correction.

4.1 Numerical Setup and Data Generation

4.1.1 Wave Regimes

We use four wave families:

1. Stokes traveling waves;
2. Tanaka solitary waves, including isolated and multiple-wave cases;
3. JONSWAP/TMA irregular waves; and
4. Benjamin–Feir configurations with a carrier and perturbing sidebands.

The dataset contains 3,686,400 snapshots per family, or 14,745,600 in total. Figure 1 shows one training snapshot from each family.

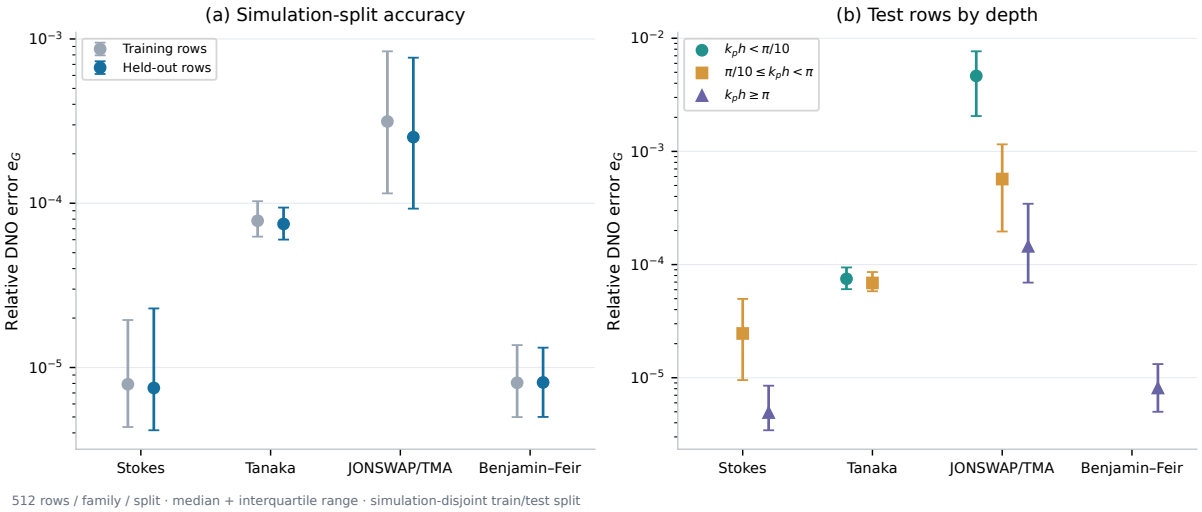


Figure 2: Relative L^2 DNO errors for 512 sampled snapshots per family per split. Left: training and test medians with interquartile ranges. Right: test errors grouped by $k_ph < \pi/10$, $\pi/10 \le k_ph < \pi$, and $k_ph \ge \pi$; empty groups are omitted.

4.1.2 Reference Solver and Dataset Construction

The test trajectories use a periodic domain of length $L = 2\pi$, $N_x = 1024$ grid points, and gravity $g = 1$. The reference DNO is computed with Craig–Sulem order $M_{\text{ref}} = 6$ and padding factor eight. Reference and learned trajectories use the same integrating-factor Gauss–Legendre scheme, with four fixed-point iterations, time step $\Delta t = 0.01$, and a spectral cutoff at mode 128. No adaptive damping is applied.

Entire simulations are assigned to training, validation, or test sets, containing 11,796,720, 1,474,440, and 1,474,440 snapshots, respectively. The one-step accuracy test samples 512 snapshots per family from each of the training and test sets.

Trajectory tests use 32 initial conditions per family from the test set, chosen across the sampled parameter ranges. We integrate to $T = 20$ for Stokes and JONSWAP/TMA and $T = 200$ for Tanaka and Benjamin–Feir, saving 251 equally spaced states per trajectory.

4.1.3 Implementation Details

The network has 1,342,400 parameters and is selected by validation error after 40 training epochs. Network evaluation uses single precision; the analytic terms, time integration, and Hamiltonian evaluation use double precision. Timing measurements use an AMD EPYC 9124 CPU with eight logical CPUs, excluding compilation.

4.2 Accuracy and Structural Properties of the Learned DNO

4.2.1 Relative DNO Accuracy

For sample j, define

$$e_G^{(j)} = \frac{\|G_\theta(\eta^{(j)}; h^{(j)})\xi^{(j)} - G(\eta^{(j)}; h^{(j)})\xi^{(j)}\|_{L^2}}{\|G(\eta^{(j)}; h^{(j)})\xi^{(j)}\|_{L^2}}. \tag{27}$$

Figure 2 compares the learned DNO with the stored reference targets, grouped by wave family and relative depth k_ph.

4.2.2 Linearity and Positive Semidefiniteness

For fixed (η, h), Dirichlet data ξ_1, ξ_2, and scalars a, b, define

$$e_{\text{lin}} = \frac{\|G_\theta(\eta;h)(a\xi_1 + b\xi_2) - aG_\theta(\eta;h)\xi_1 - bG_\theta(\eta;h)\xi_2\|_{L^2}}{|a|\|G_\theta(\eta;h)\xi_1\|_{L^2} + |b|\|G_\theta(\eta;h)\xi_2\|_{L^2}}. \tag{28}$$

We perform 100 independent trials per family. The surface-depth pair and two Dirichlet inputs are drawn independently from training snapshots of that family; the inputs are centered, and a, b are sampled independently from $\mathcal{U}(-1, 1)$. Mean linearity residuals range from 3.37×10^{-6} to 1.50×10^{-4} across the four families with single-precision network evaluation.

Because constant Dirichlet data belong to the DNO nullspace, its quadratic form is positive semidefinite on the full space. For $\widetilde{\xi} = \xi - L^{-1}\int_0^L \xi(x)\,dx$, define

$$q_\theta(\eta, \widetilde{\xi}; h) = \frac{\langle \widetilde{\xi}, G_\theta(\eta;h)\widetilde{\xi}\rangle_{L^2}}{\|\widetilde{\xi}\|_{L^2}^2}. \tag{29}$$

On 100 independently sampled matched training snapshots per family, no quadratic form falls below $-\tau$, with $\tau = 10^{-10}$. This finite sample test is not a proof of positive semidefiniteness of the learned operator.

4.3 Trajectory Accuracy and Long-Time Stability

Define the trajectory errors by

$$e_\eta(t) = \frac{\|\eta_\theta(t) - \eta_{\text{CS}}(t)\|_{L^2}}{\|\eta_{\text{CS}}(t)\|_{L^2}}, \qquad e_\xi(t) = \frac{\|\xi_\theta(t) - \xi_{\text{CS}}(t)\|_{L^2}}{\|\xi_{\text{CS}}(t)\|_{L^2}}. \tag{30}$$

Potential fields are compared in the mean-zero gauge. For saved times $t_0, \ldots, t_{N_t}$, the surface trajectory error is

$$E_{\eta,\text{traj}} = \left(\frac{\sum_{n=0}^{N_t}\|\eta_\theta(t_n) - \eta_{\text{CS}}(t_n)\|_{L^2}^2}{\sum_{n=0}^{N_t}\|\eta_{\text{CS}}(t_n)\|_{L^2}^2}\right)^{1/2}, \tag{31}$$

with $E_{\xi,\text{traj}}$ defined analogously. The terminal errors are $e_\eta(T)$ and $e_\xi(T)$.

4.3.1 Overall Trajectory Accuracy and Stability

Figure 3 summarizes all 128 test trajectories. All saved states are finite and have positive fluid depth.

4.3.2 Regime-Specific Physical Accuracy

Traveling Stokes and Tanaka Waves For a periodic traveling wave, define the translation error by

$$\Delta x(t) = \operatorname*{argmin}_{s\in[-L/2, L/2)} \|\eta_\theta(\cdot, t) - \eta_{\text{CS}}(\cdot - s, t)\|_{L^2}. \tag{32}$$

The translation-aligned profile error is

$$e_{\eta,\text{align}}(t) = \frac{\|\eta_\theta(\cdot, t) - \eta_{\text{CS}}(\cdot - \Delta x(t), t)\|_{L^2}}{\|\eta_{\text{CS}}(\cdot, t)\|_{L^2}}. \tag{33}$$

Figure 4 separates phase drift from profile distortion for Stokes and single-wave Tanaka cases near their respective median terminal errors. Alignment uses Fourier interpolation; Stokes displacements are unwrapped modulo one carrier wavelength.

Tanaka-Wave Interactions For the intended collision study, define the collision time as the time of minimum periodic separation between two tracked crests, and set

$$e_{\text{coll}} = \frac{|T_{\text{coll},\theta} - T_{\text{coll,CS}}|}{T_{\text{coll,CS}}}. \tag{34}$$

Figure 5 shows profiles around a local maximum of reference crest amplification. Both solutions use the same fixed spatial shift. The tracked-crest collision-time error in (34) is not measured here.

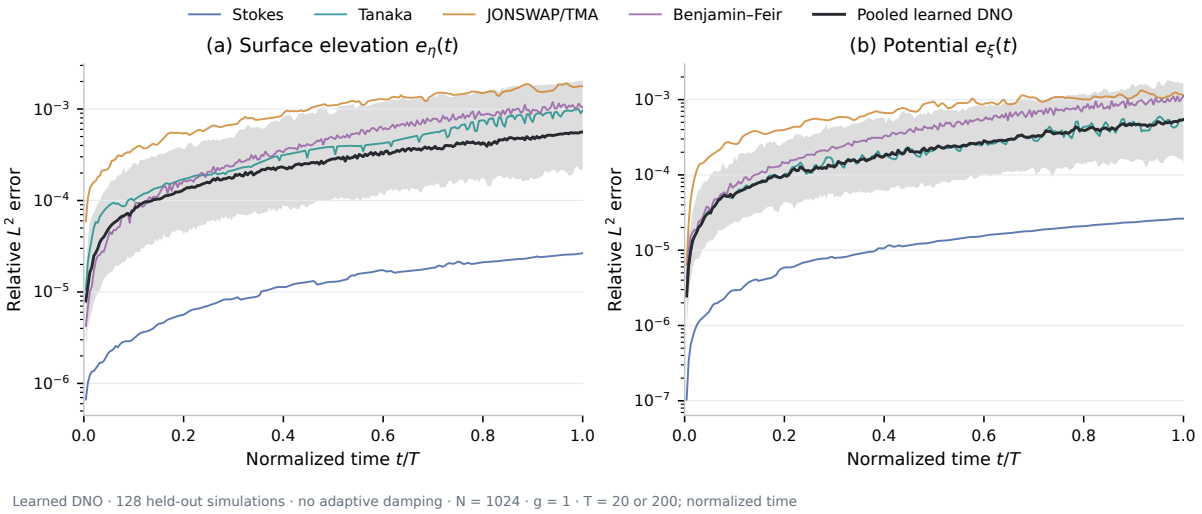


Figure 3: Relative surface and potential errors for 128 test trajectories, 32 per family. Black curves and shading show the overall median and interquartile range; colored curves show family medians. Time is normalized by each family's integration horizon T.

Benjamin–Feir and JONSWAP/TMA Waves For periodic wavenumbers $k = 2\pi n/L$, define

$$\lambda_k(h) = |k| \tanh(h|k|), \qquad \mathcal{E}_k(t) = \frac{1}{2}\left[g|\widehat{\eta}_k(t)|^2 + \lambda_k(h)|\widehat{\xi}_k(t)|^2\right]. \tag{35}$$

The normalized spectral-energy distribution and its transfer from the initial state are

$$p_k(t) = \frac{\mathcal{E}_k(t)}{\sum_j \mathcal{E}_j(t)}, \qquad \Delta p_k(t) = p_k(t) - p_k(0). \tag{36}$$

For a carrier k_0 with symmetric sidebands spaced by q, define

$$R_{\mathrm{sb}}(t) = \frac{\mathcal{E}_{k_0-q}(t) + \mathcal{E}_{k_0+q}(t)}{\mathcal{E}_{k_0}(t)}. \tag{37}$$

Figure 6 shows two held-out examples. For Benjamin–Feir waves, we use the strongest symmetric sideband pair in the initial spectrum. For JONSWAP/TMA, the initial peak mode n_p defines low ($0 < n < 0.5n_p$), peak ($0.5n_p \le n \le 1.5n_p$), and high ($n > 1.5n_p$) bands. One-sided spectra include both members of each conjugate-mode pair.

4.4 Hamiltonian Conservation

To assess physical energy independently of the learned DNO, we evaluate the Hamiltonian of a predicted state using the reference Craig–Sulem operator:

$$H^\theta_{\mathrm{ref}}(t) = \frac{1}{2}\int_\Omega \left[\xi_\theta(t) G(\eta_\theta(t); h)\xi_\theta(t) + g\eta_\theta(t)^2\right] dx. \tag{38}$$

The signed and absolute relative drifts are

$$\delta_H(t) = \frac{H^\theta_{\mathrm{ref}}(t) - H^\theta_{\mathrm{ref}}(0)}{|H^\theta_{\mathrm{ref}}(0)|}, \qquad e_H(t) = |\delta_H(t)|. \tag{39}$$

We evaluate the sixth-order reference DNO on both solutions at 26 times per trajectory. Figure 7 shows the resulting drift, which is larger for the learned trajectories than for the reference.

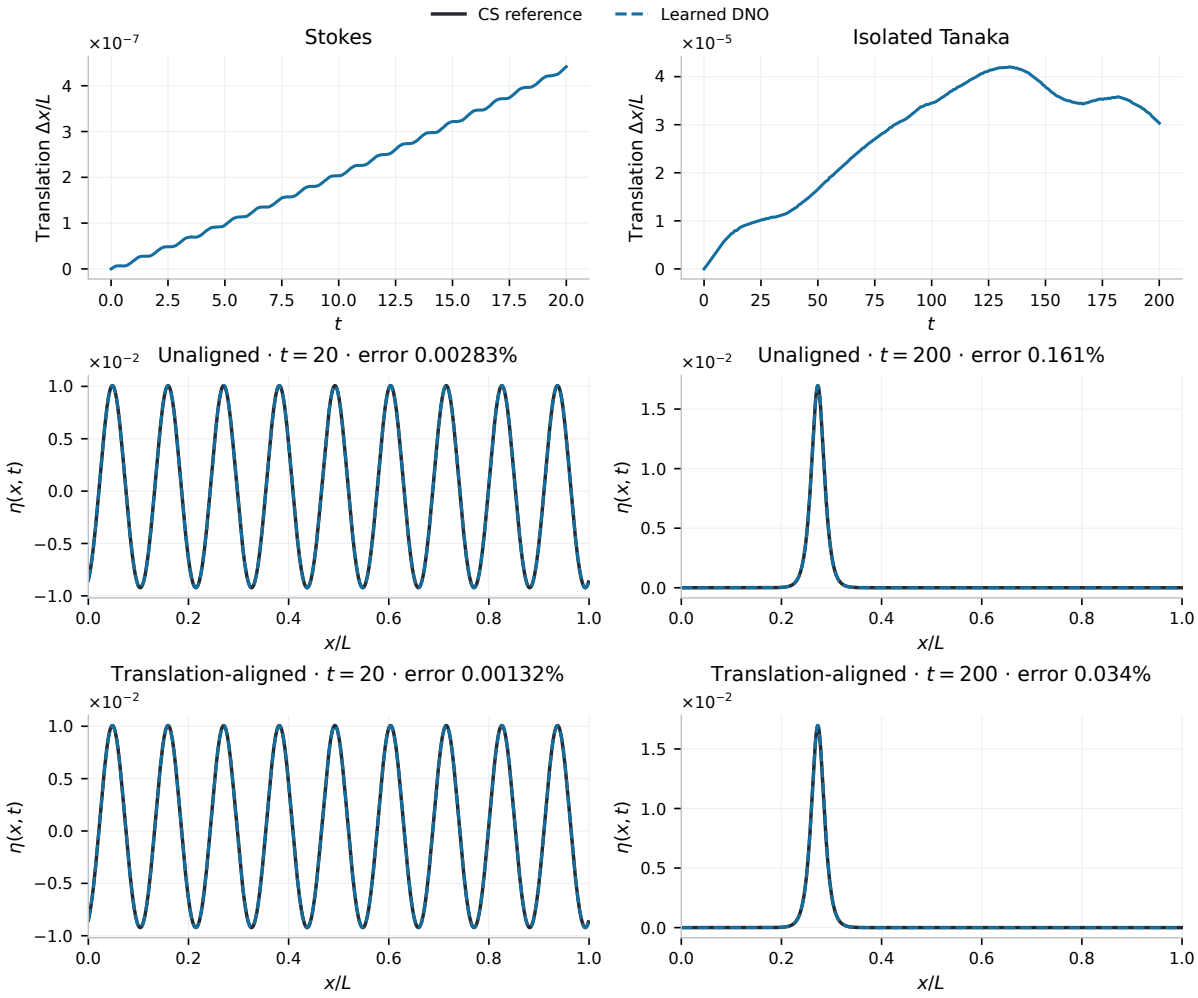


Figure 4: Traveling Stokes and Tanaka waves. Top: displacement error $\Delta x/L$. Middle: reference and learned profiles. Bottom: profiles after translation alignment. Both unaligned and aligned errors are reported.

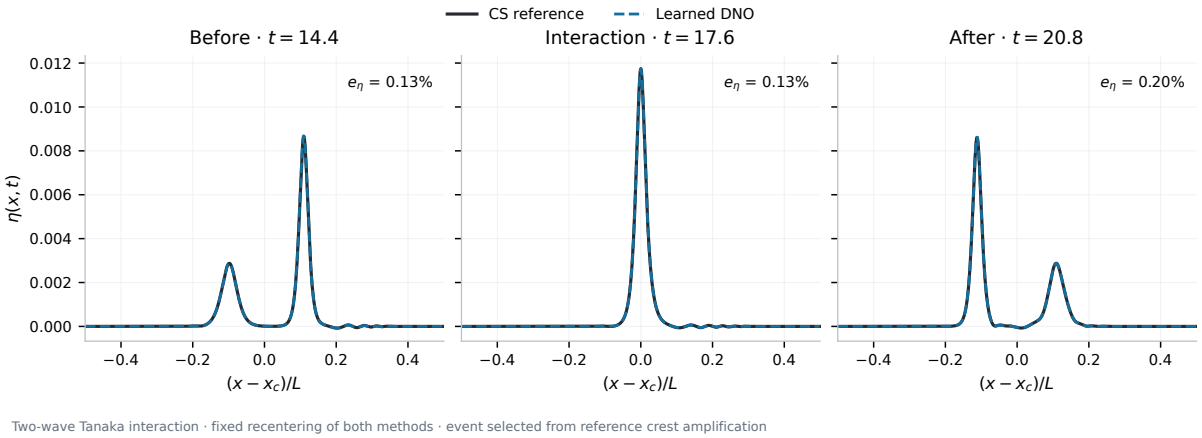


Figure 5: A two-wave Tanaka interaction at depth $h = 0.039392$. Reference and learned profiles are shown on identical axes at $t = 14.4$, 17.6, and 20.8, with relative surface errors.

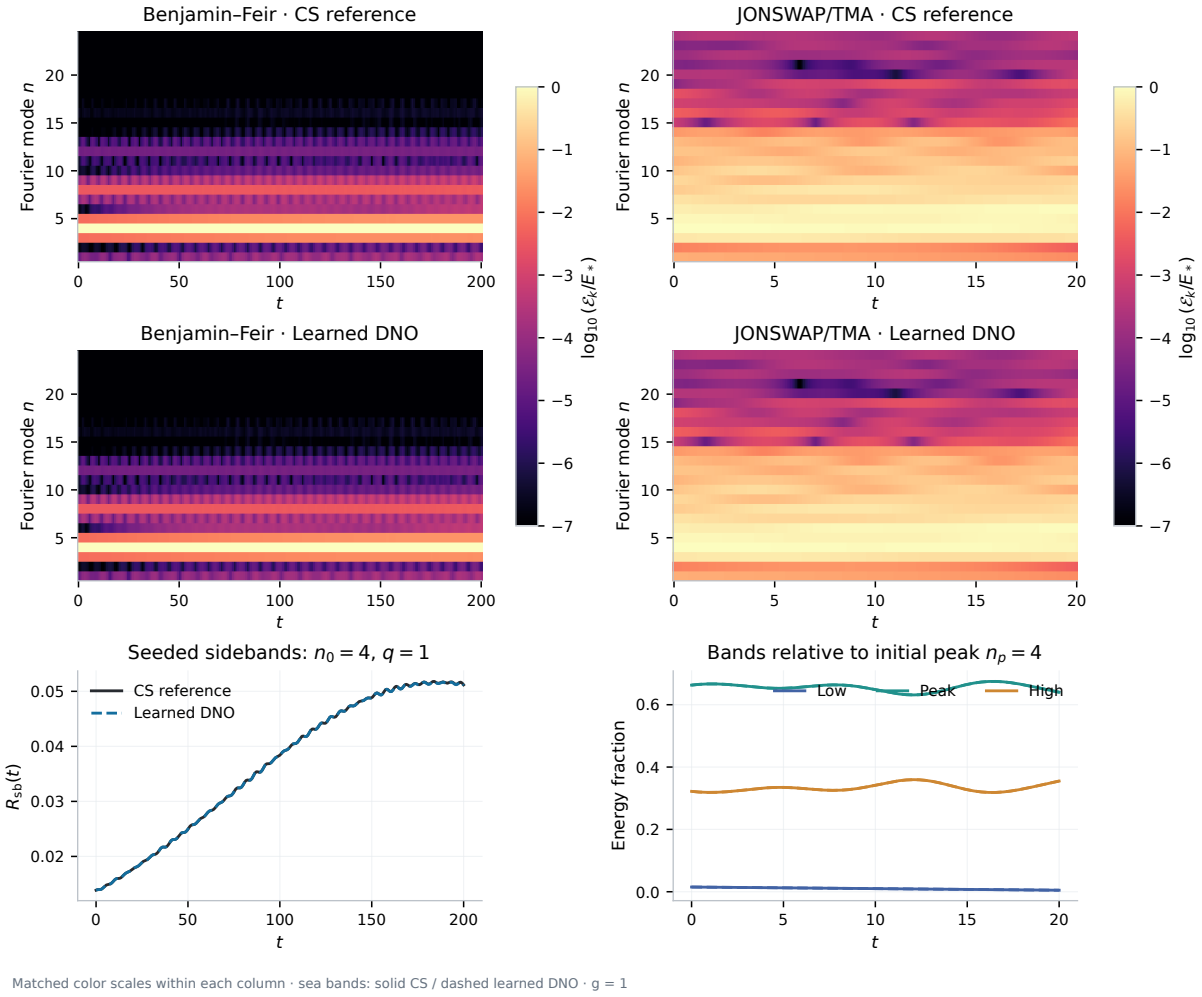


Figure 6: Spectral energy for Benjamin–Feir and JONSWAP/TMA waves. Paired heatmaps share a color scale normalized by their joint maximum E_*. Below: sideband-to-carrier ratio and band-energy fractions. Solid curves denote the reference; dashed curves denote the learned DNO.

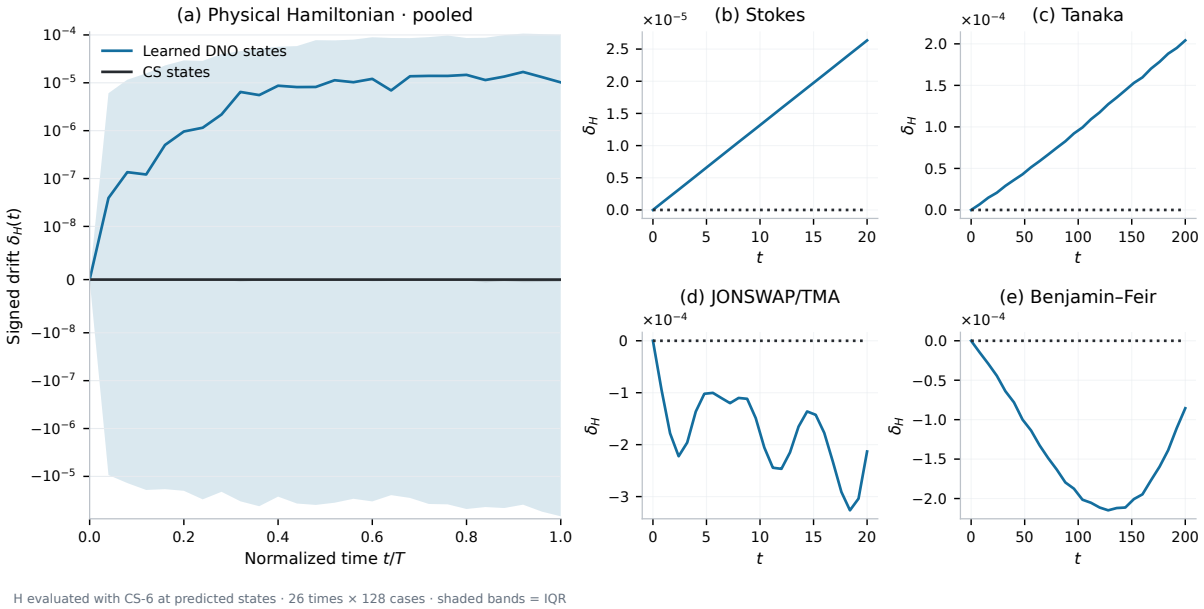


Figure 7: Relative Hamiltonian drift evaluated with the reference DNO. Medians and interquartile ranges over 128 test cases use a symmetric-log scale with linear threshold 10^{-8}; the four representative cases use linear axes.

4.5 M-Independent Evaluation and Computational Cost

4.5.1 FNO Complexity and Independence from M

For a Craig–Sulem expansion of order M, the cost of one DNO evaluation is

$$C_{\mathrm{CS}}(N_x, M) = \Theta(M^2 N_x \log N_x). \tag{40}$$

For a one-dimensional FNO-type model with L_θ Fourier layers, channel width C, and K retained modes, a representative evaluation cost is

$$C_{\mathrm{FNO}}(N_x) = \mathcal{O}\big(L_\theta[CN_x \log N_x + N_x C^2 + KC^2]\big)\,. \tag{41}$$

For fixed L_θ, C, K, the FNO cost is $\mathcal{O}(N_x \log N_x)$, independent of M; this alone does not imply a wall-clock speedup. Specialize this expression to the network architecture.

Figure 8 times one DNO evaluation for a held-out Stokes wave, Fourier-resampled to each grid. Timings exclude compilation and use seven synchronized repeats.

4.5.2 Fixed-Horizon Trajectory Runtime

For Figure 9, we integrate one held-out Stokes wave at $N_x = 256$ to $T = 1$, with $\Delta t = 0.01$ and a mode-64 cutoff. Both methods use the same Gauss–Legendre integrator, four fixed-point iterations, padding factor eight, and 51 saved states. Timings are medians of three synchronized runs after compilation. The learned DNO is slower than the reference series in this CPU experiment.

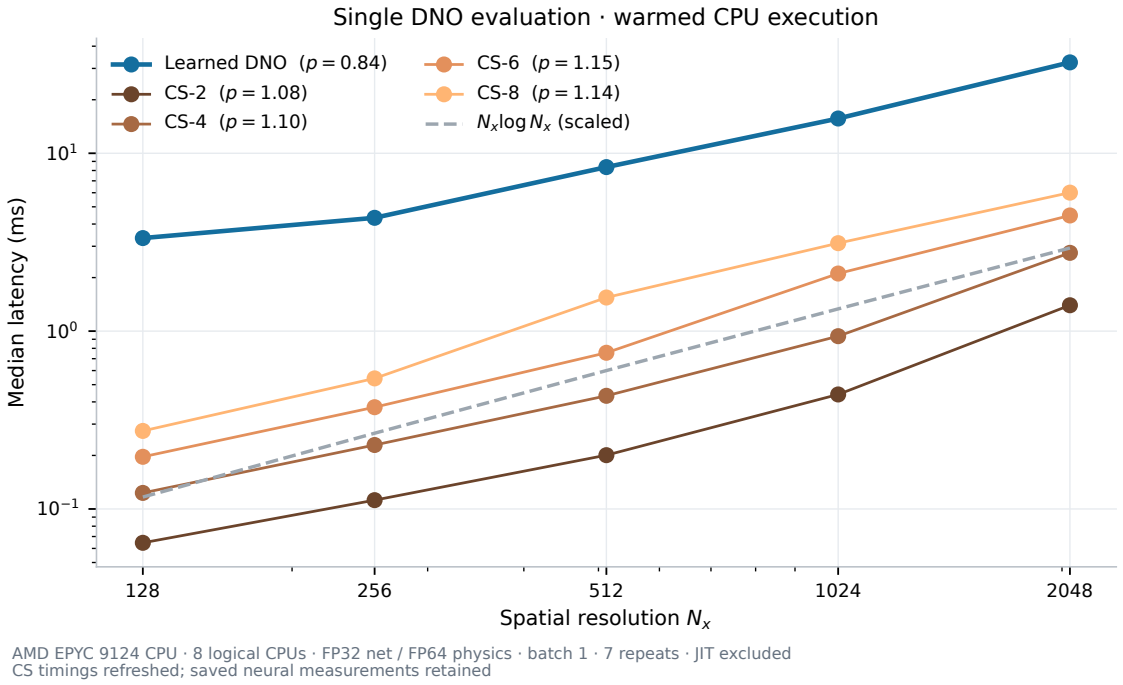


Figure 8: CPU time per DNO evaluation for the learned operator and Craig–Sulem orders 2, 4, 6, and 8. Curves show median times over seven repeats; fitted slopes describe the measured resolution range. A scaled $N_x \log N_x$ curve is shown for comparison.

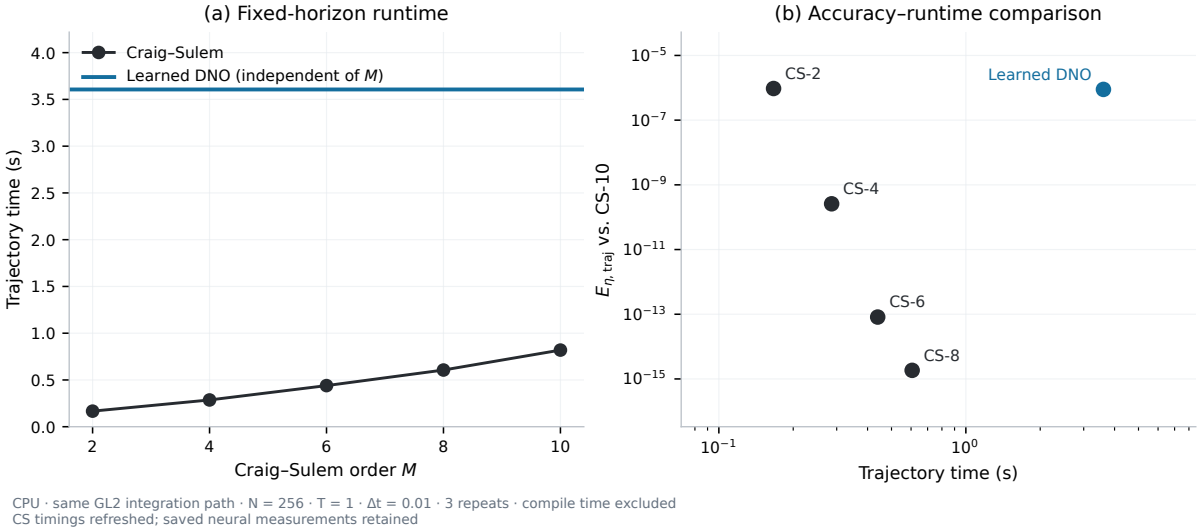


Figure 9: CPU runtime and accuracy for a Stokes trajectory at $N_x = 256$, $T = 1$, and $\Delta t = 0.01$. Left: runtime versus series order, with the learned DNO shown for comparison. Right: surface trajectory error relative to the order-10 reference versus runtime; its zero self-error is omitted from the logarithmic plot.

5 Conclusion and Discussion

Limitations Complete after the remaining comparisons.

Future Work

Acknowledgments

This work was supported in part by...

References

Use the project's references.bib when the citations are completed. No bibliography file is needed to view this draft.

A Time-stepping Modal Error

We demonstrate that small errors in the Fourier modes of a learned neural surrogate for the DNO G cause long-time accumulation of modal phase error under our symplectic integration scheme.

B Low-Rankedness of Nonlinear Residuals of CS

C More About the Dataset